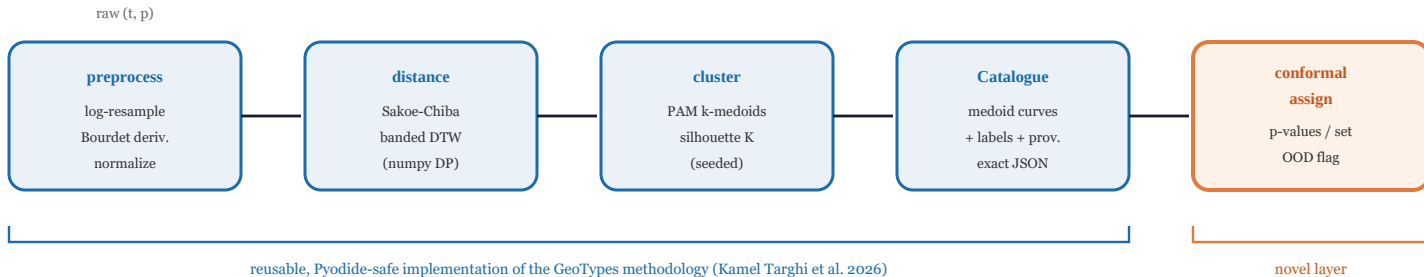


pygeotypes: from response signals to a catalogue with guarantees



class-conditional split-conformal:

$p_g(x) = (\#\{ s \text{ in } S_g : s \geq d(x, \text{medoid}_g) \} + 1) / (|S_g| + 1)$, with nonconformity score $d = \text{DTW distance to medoid } g$.

prediction set = $\{ g : p_g > \alpha \}$ covers the true GeoType with probability $\geq 1 - \alpha$ per class (Vovk et al. 2005; Angelopoulos & Bates 2023).

An empty set is an honest out-of-catalogue flag: the shape matches no behaviour class at confidence $1 - \alpha$.